



Underfitting and Overfitting



Course Objectives

The implications of underfitting and overfitting

The causes of underfitting and overfitting

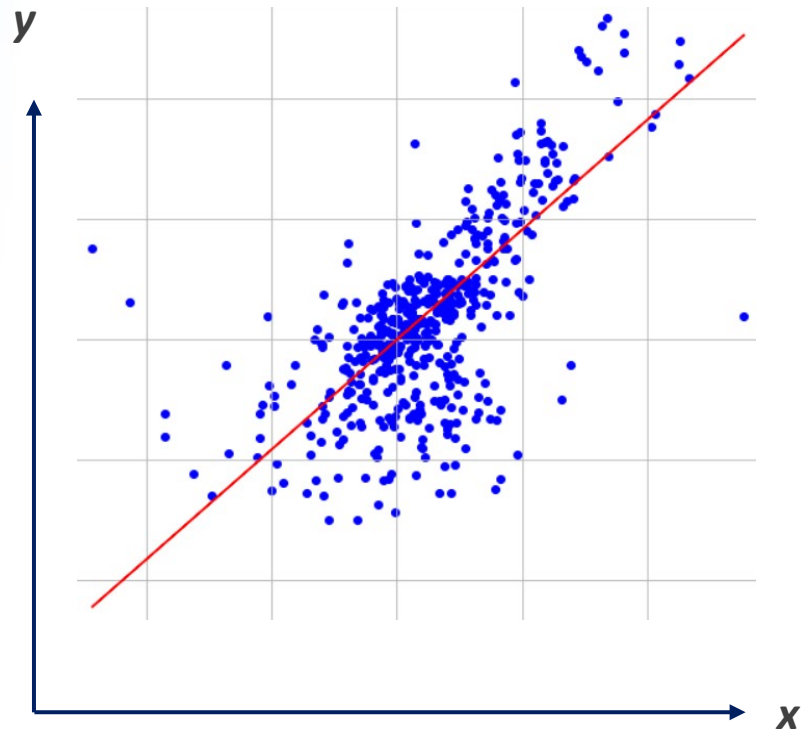
The approaches to handle underfitting and overfitting

Learning Outcomes

At the end of the course, you will be able to

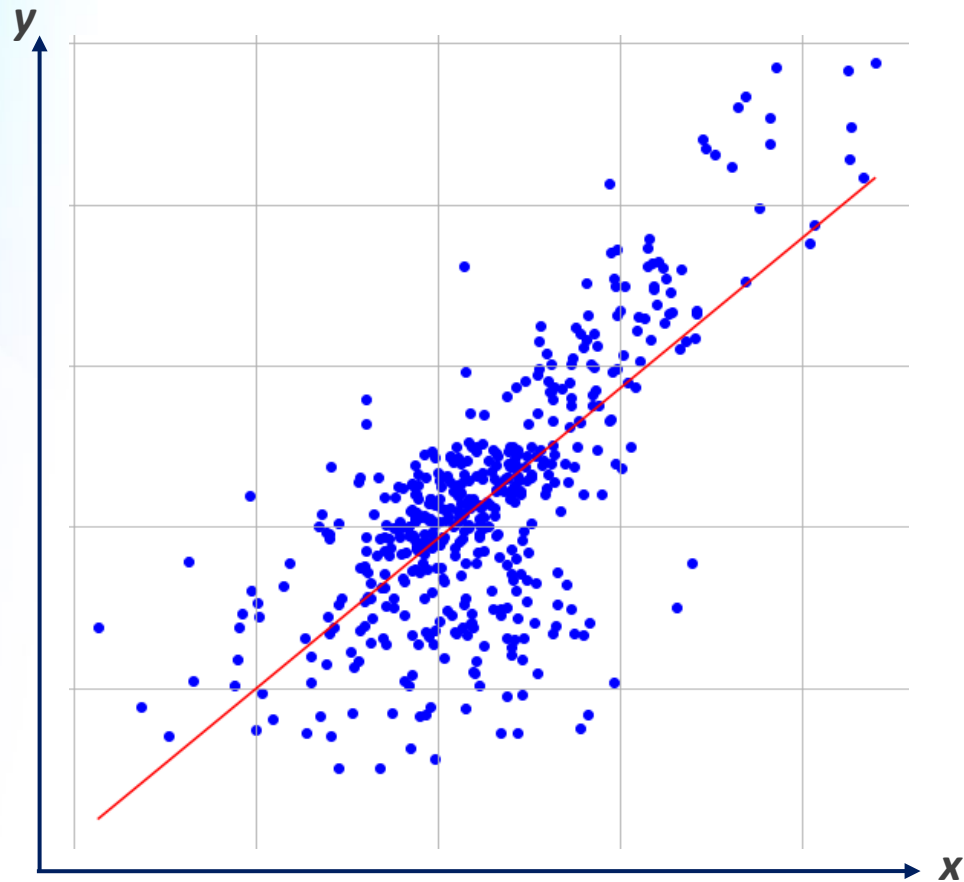
- Understand what are underfitting and overfitting
- The causes of underfitting and overfitting
- The approaches to prevent underfitting and overfitting

An Example

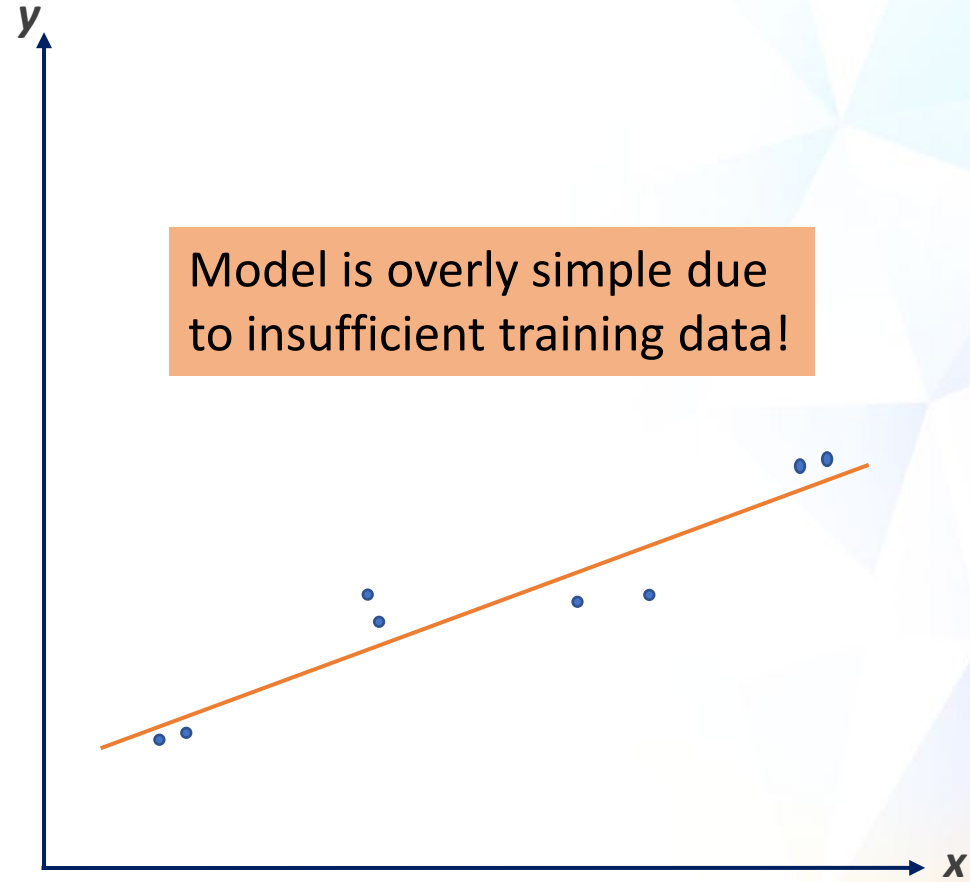


Good sample of data

Underfitting and Causes



Good sample of data

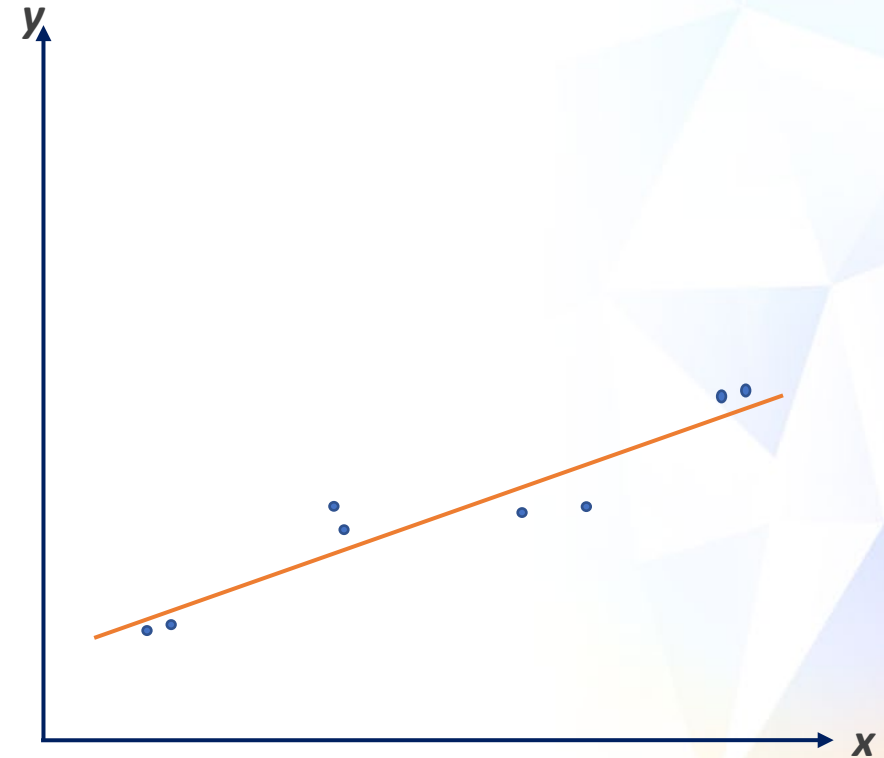


Model is overly simple due to insufficient training data!

Insufficient training data

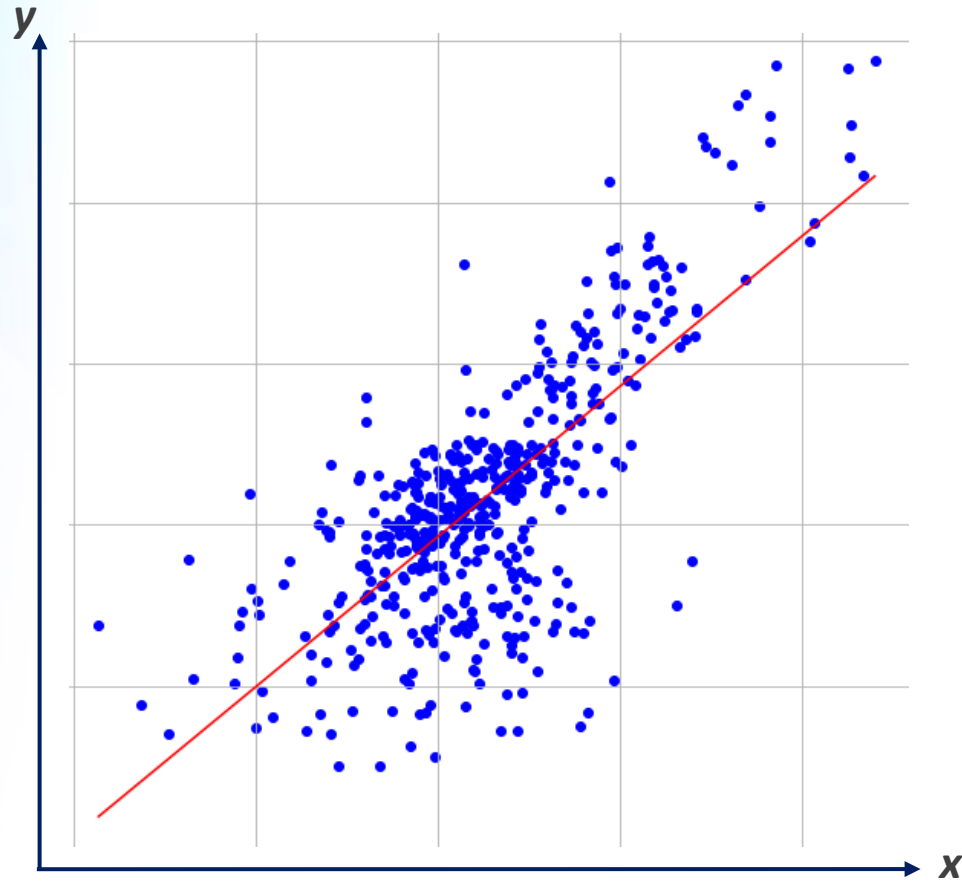
Addressing Underfitting

- Approaches
 - Add features
 - Add data

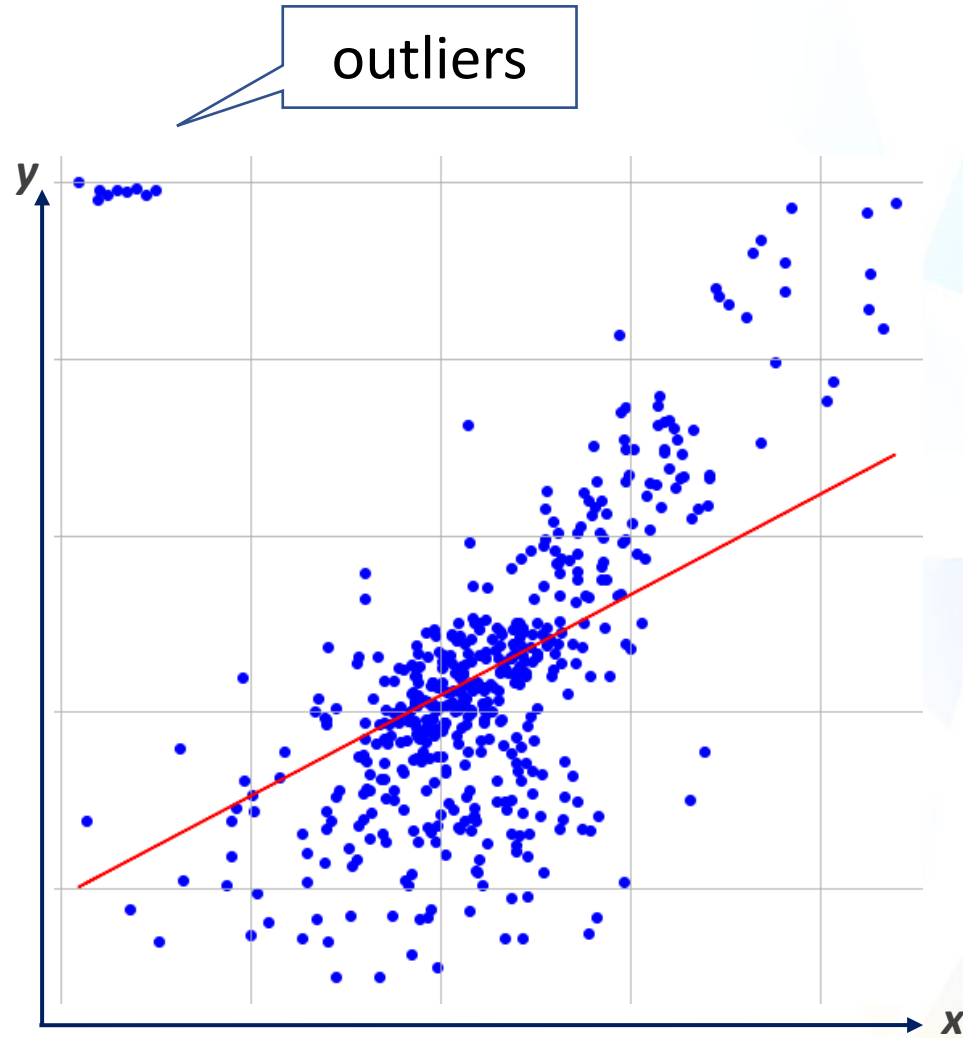


Insufficient training data

Overfitting



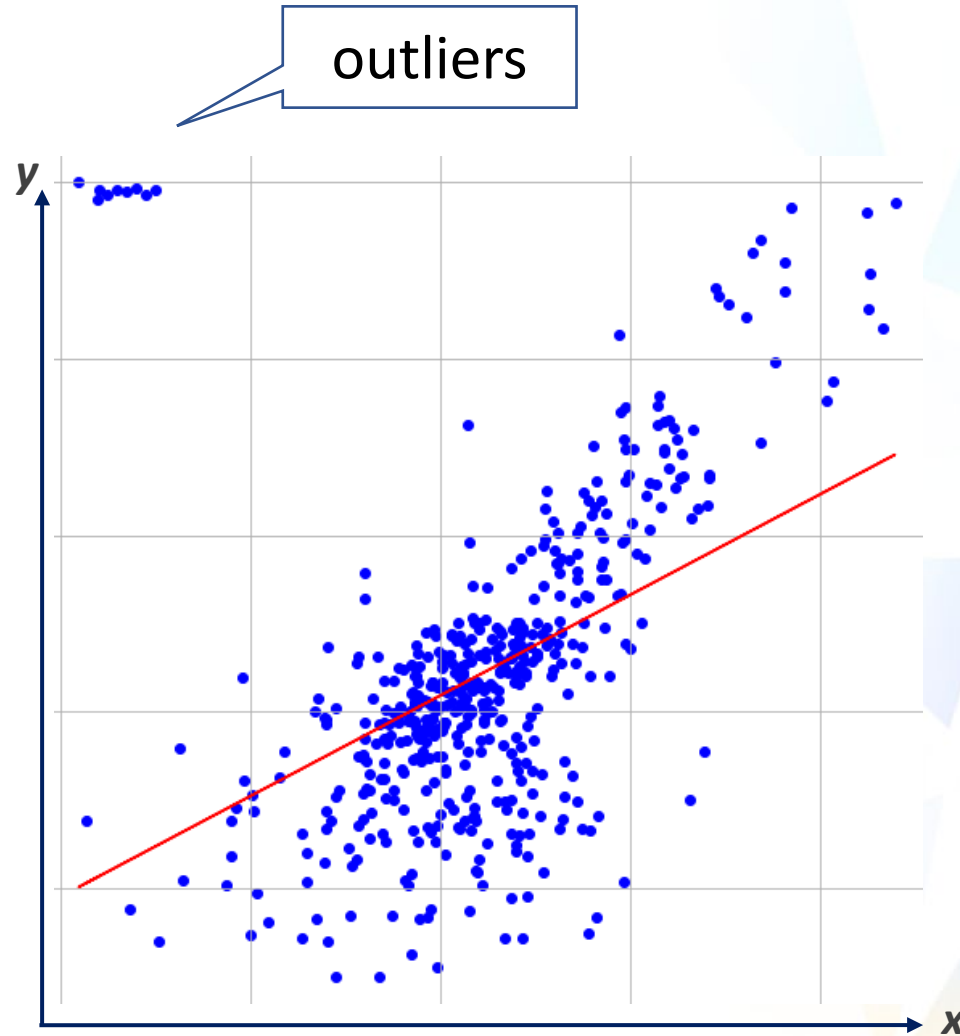
Good sample of data



Poor representation of data

Overfitting

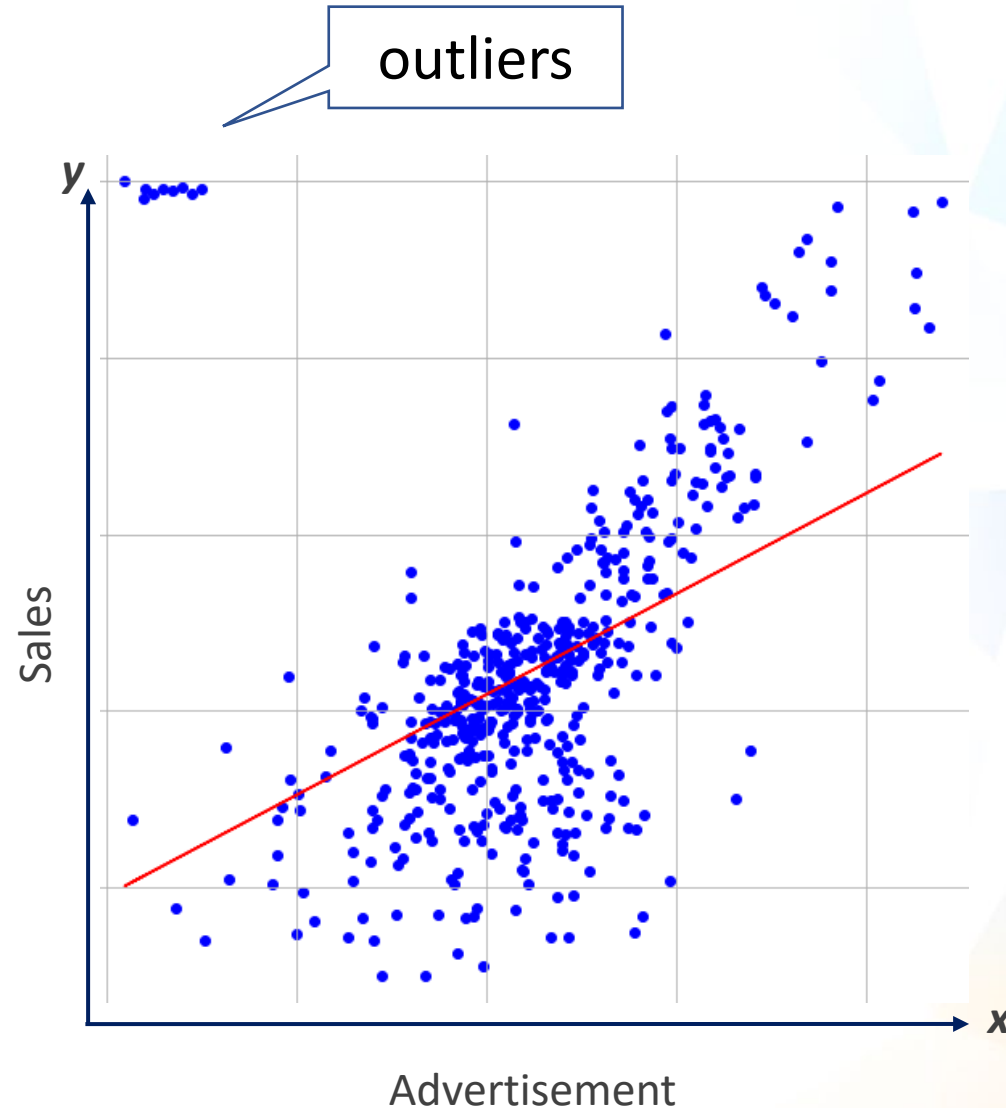
- overlearning (or overfitting) is more difficult to detect and correct
- noise or random fluctuations in training data are perceived by the model



Outliers

Datasets have outliers

- Strive towards understanding the **general patterns** that emerge from these data points
 - *instead of fitting to every single one of them*



Addressing Overfitting

- Removal of outliers
- Fewer features
- Cross validation
- Starts with simpler features

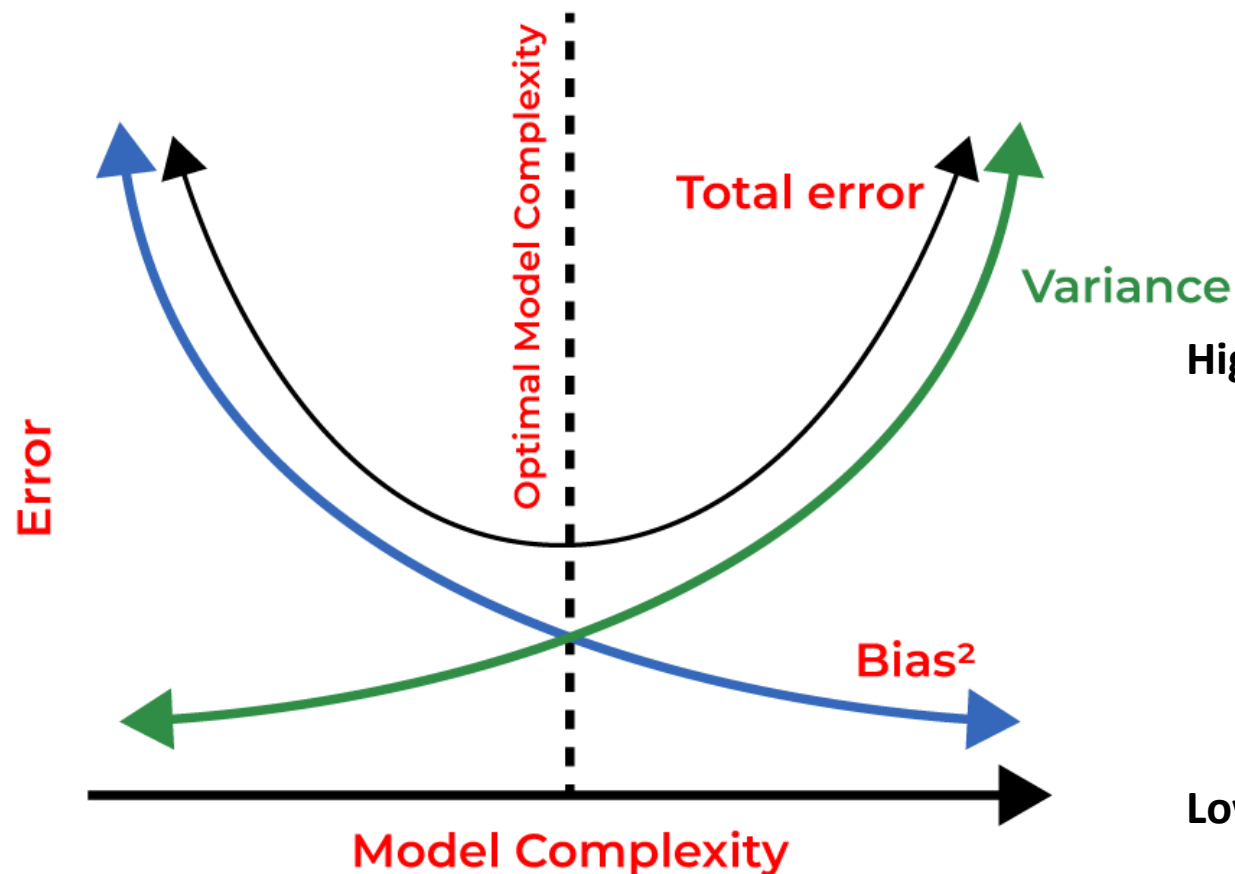
What Have We Learnt?

- Underfitting
 - What it is
 - The corrective actions
- Overfitting
 - What it is
 - The corrective actions

Underfitting and Overfitting

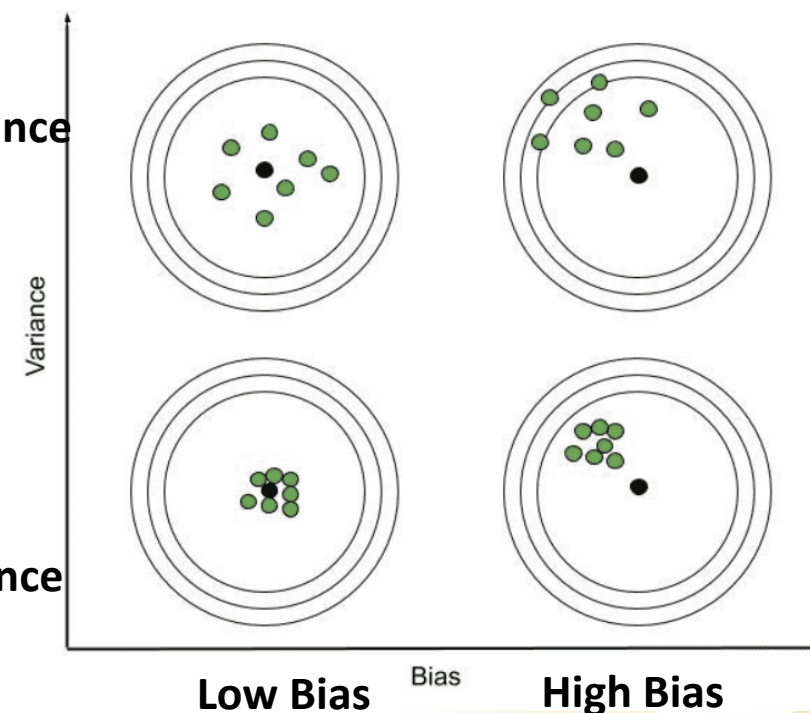
Underfitting -> High Bias and Low Variance

Overfitting -> Low Bias and High Variance

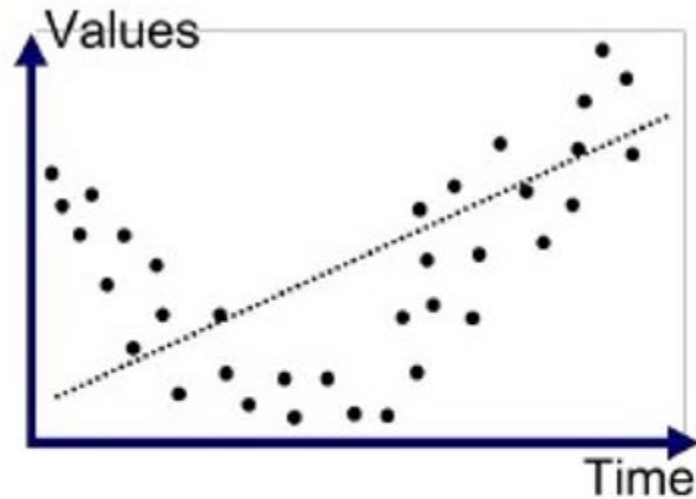


High Variance

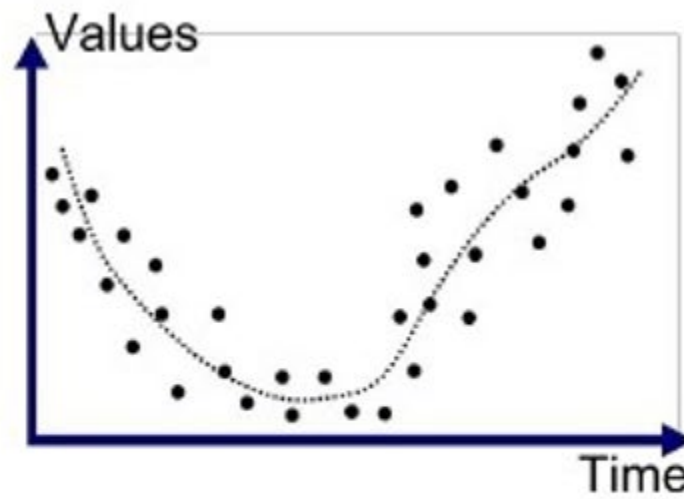
Low Variance



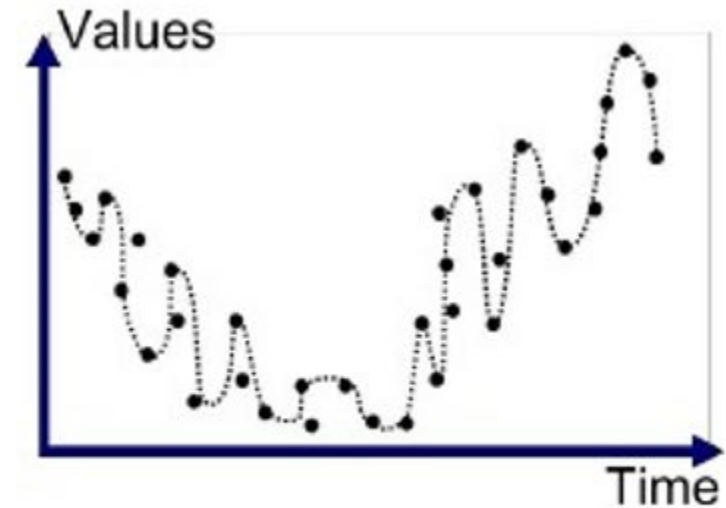
Bias vs Variance



Underfitted



Good Fit/Robust



Overfitted

- **Bias** represents the difference between the average prediction of our model and the correct value we are trying to predict.
- **Variance** quantifies the model's sensitivity to small fluctuations or changes in the training data. In other words, it is the variability of the model that how much it is sensitive to another subset of the training dataset.

Interesting Articles

1. <https://towardsdatascience.com/overfitting-vs-underfitting-a-complete-example-d05dd7e19765>
2. https://scikit-learn.org/stable/modules/linear_model.html
3. <https://www.analyticssteps.com/blogs/l2-and-l1-regularization-machine-learning>
4. <https://towardsdatascience.com/dont-sweat-the-solver-stuff-aea7cddc3451>
5. <https://www.geeksforgeeks.org/bias-vs-variance-in-machine-learning/>